Part A: Data Processing

(10 points)

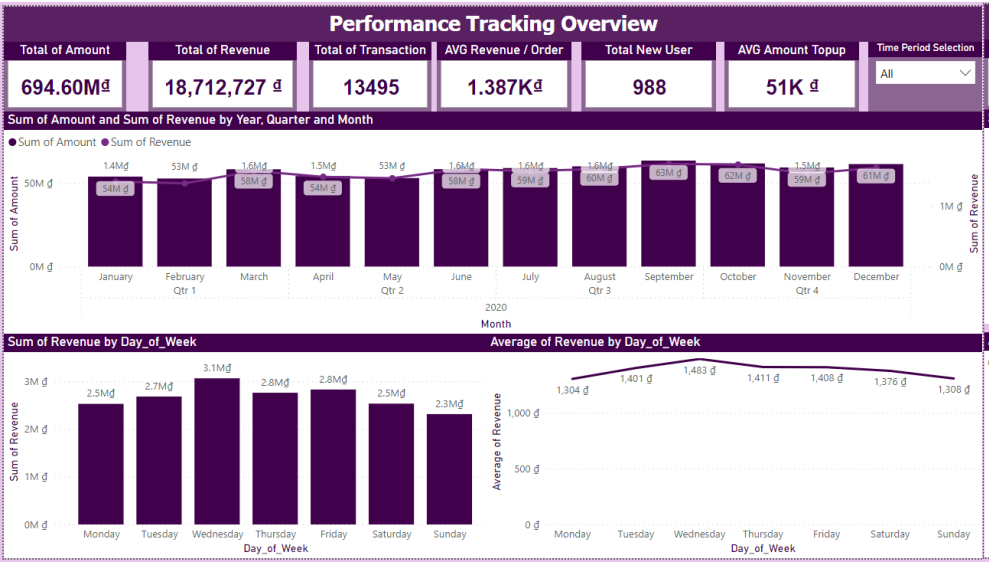
Part A

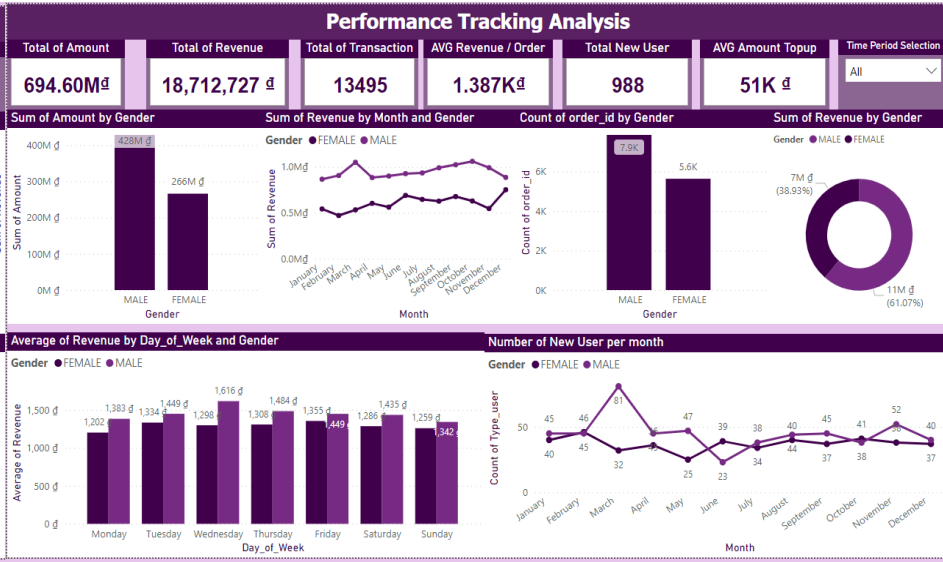
Question	Answer	Excel file data location (sheet/cell)
1) Total revenue in January 2020:	1 409 827	Python notebook / Dashboard
2) Most profitable month:	2020-09: 1 702 200	Python notebook / Dashboard
3) Day of week (most money):	Wednesday', 1482.535074987905	Python notebook / Dashboard
3) Day of week (least money):	'Monday', 1304.437564499484	Python notebook / Dashboard
4) Total number of New users in Dec 2020:	76	Python notebook / Dashboard

Part B: Data Visualization

(10 points)

Part B: Q5: Performance Dashboard





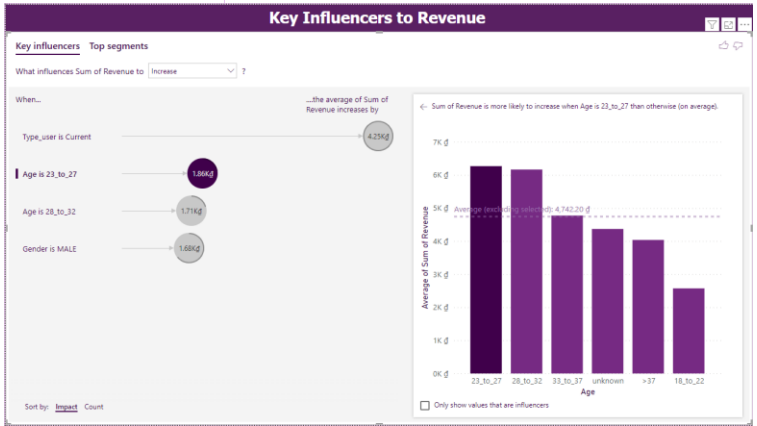
Part B: Q5: Performance Dashboard

Decomposition Tree of Revenue Stream

Merchant_name: Viettel | Type_user: Current | Gender: MALE | Age: 23_to_27 | Day_of_Week

- **Sum of Revenue** 18,712,727 ₫
  - Gmobile 3,200 ₫
  - Mobifone 5,757,127 ₫
  - Vietnamobile 867,200 ₫
  - **Viettel** 7,152,400 ₫
    - **Current** 6,659,400 ₫
      - FEMALE 2,396,200 ₫
      - **MALE** 4,263,200 ₫
        - >37 364,000 ₫
        - 18_to_22 379,200 ₫
        - **23_to_27** 1,155,600 ₫
          - Monday 183,200 ₫
          - Tuesday 160,000 ₫
          - Wednesday 153,400 ₫
          - Thursday 151,000 ₫
          - Friday 165,200 ₫
          - Saturday 180,400 ₫
          - Sunday 162,400 ₫
        - 28_to_32 1,003,600 ₫
        - 33_to_37 562,400 ₫
        - unknown 798,400 ₫
    - New 493,000 ₫
  - Vinaphone 4,932,800 ₫

Key Influencers to Revenue



Part C: Analysis & Business Insights

(10 points)

Part C: Q6



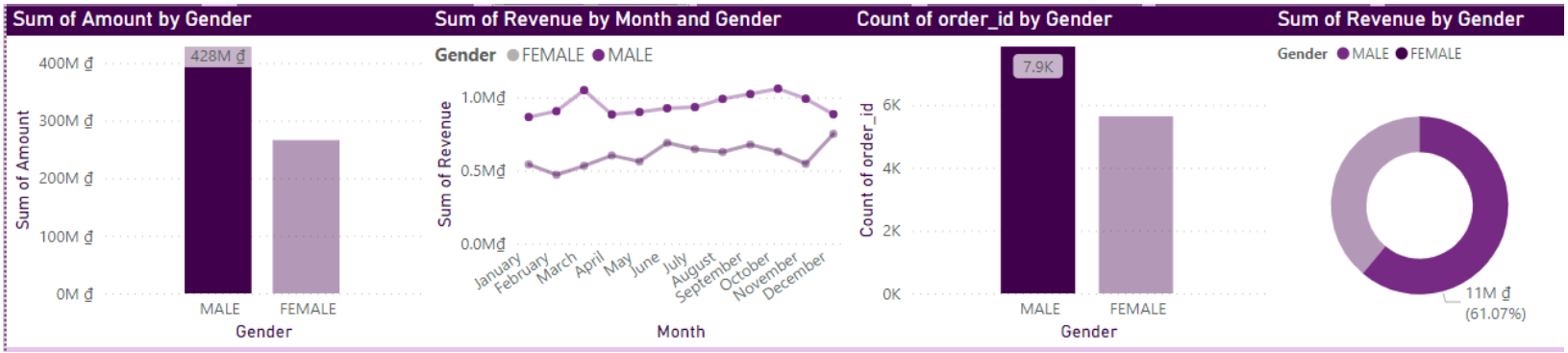
- The middle of the week (Wednesday) has the highest profit at $3,064,400.

- The beginning (Monday, Tuesday) and the end (Friday) of the week also show higher revenues compared to weekends.

- Revenue tends to dip slightly on weekends, with Sunday being the lowest at $2,313,927.

Part C: Q6

Males have generated a higher total revenue ($11,427,100) compared to females ($7,285,627) in perspective 61.07% in total. Males also makes a higher number of transactions than females.





- 'Viettel' is the leading merchant in terms of revenue generation with a total of $7,152,400.
- 'Mobifone' and 'Vinaphone' are also significant contributors with $5,757,127 and $4,932,800, respectively.
- 'Vietnamobile' and 'Gmobile' have much lower total revenues of $867,200 and $3,200.

Part C: Q7

Recommendation actions for revenue optimization:

• Targeted campaigns towards male users as they contribute more to the revenue.

• Collaborations or promotions with the top merchants like Viettel, Mobifone, and Vinaphone, which are more profitable.

• Special attention to midweek days for pushing promotions or offers, new products to capture more on the higher revenue trends and interactions and for up/cross sell opportunities as topping up in the Apps for this peak time.

• Exploring why weekends have a drop in revenue and considering strategies to boost sales during these periods.

• Current user: boosts cashback or loyalty program for the top up 50k due to the average spending is 52k to retain them. And boost the conversion of new user around 30k top up due to their average spending is 32k.

Part C: Q8

Financial Impact:
Higher cashback to user, lower revenue to Momo. This is the tradeoff for the business. This tradeoff potentially gains for Momo for **increased user engagement** and **higher transaction volumes** that could offset the reduced revenue.

User Engagement:
Higher cashback rates can enhance user engagement. They may use MoMo more frequently and could potentially shift their spending from competitors to MoMo.

Competitive Positioning:
By offering higher cashback rates, MoMo could differentiate itself from competitors. This could be particularly effective if the rates are higher than what competitors offer or if they apply to merchants where MoMo has a strong user base.

User Acquisition and Retention:
Increased cashback could serve as an incentive for new user acquisition and current user retention. They could lead to self-promoting among users through their network. This could be a part of budget for marketing for specific time periods of the year to reach the KPIs on users growth and retention.

Part C: Q8

Merchant Relations:
Merchants may respond positively if the increased cashback leads to more transactions. Conversely, they could be concerned about the potential for reduced profit margins if they're required to contribute to the cashback scheme.

Break-even Analysis:
A break-even analysis would be necessary to determine how much additional spending is required per user to offset the increased cost of cashback.

In conclusion, Momo should conduct a detailed financial analysis on the revenue reduced on the downside and for the upside a detailed KPI metrics Momo wants to reach on applying this new scheme cashback. The starting of testing for a limited-time for this scheme to gauge the actual impact on increasing user behaviors, engagement, transactions volume, new ones and merchants' interactions before implementing on a big scale

Part C: Q9

Loyalty Programs:

1.Tiered Rewards System: user earn points as topping up and use the points for other payments and rewards.
2.Milestone Rewards: Offer bonuses when a user reaches a certain spending threshold on top-ups within a time frame.

Partnerships and Collaborations:

1. Co-branded Partnerships: Partner with merchantsto offer exclusive deals for MoMo users, such as bonus data or minutes.
2. Cross-Promotion with Other Services: Bundle top-up services with other popular transactions on MoMo to increase the perceived value, like movie tickets or ride-hailing services.

Community Engagement:

1. Social Responsibility: Launch a program where a portion of top-up fees goes to social causes, aligning with the values of society situations or Momo's community strategy.
2. Gamification: Integrate gamification elements where top-ups can earn users badges, levels, or spots on leaderboards.

THANK YOU